

ANALOGUE CLOCK WITH ADD-ON CALENDAR

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Numerous dedicated ICs are available for building digital clocks. However, the hobbyists have very limited choice for adding functions to digital clock circuits and also the dedicated ICs are costly.

Try assembling an analogue (look-alike) clock with calendar using basic digital ICs. It has a totally different look and is a lot of fun. Here we present such an analogue display clock circuit with add-on calendar. Individual sections of this circuit can be built separately and assembled

together. Hobbyists can experiment adding several different features to the circuit.

Any discarded wall clock can be used for the purpose. Hours are displayed through big (10mm) red LEDs and minutes are displayed through small (5mm) green LEDs. Seconds are displayed through two 7-segment displays at the centre of the clock, instead of 60 LEDs to display a moving seconds pointer. This minimises the cumbersome wiring and component count. Also, it gives a novel look to the clock. Date, day, month and year display is optional. 6V battery backup supply ensures time display without interruption, maintaining the correct time.

Circuit description

Fig. 1 shows the block diagram of the analogue display clock. It is made up of following basic circuits put together to function as a clock with calendar display:

1. 1Hz clock generator circuit
2. Seconds counter and display circuit
3. Minutes counter and display circuit
4. Hours counter and display circuit
5. Day and date counter and display circuit
6. Month and year display circuit

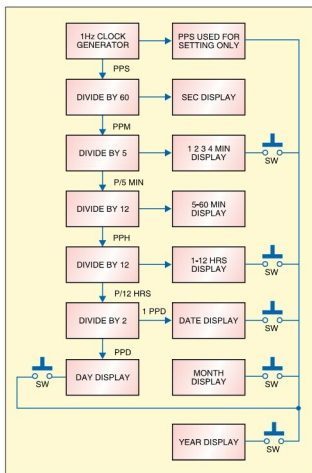


Fig. 1: Block diagram for analogue clock with add-on calendar



The proposed analogue clock panel along with LEDs and switches is shown in Fig. 2 and the power supply circuit for the analogue clock in Fig. 3. Fig. 4 shows the main circuit for the analogue clock with add-on calendar.

1Hz clock generator. The clock generator circuit uses IC CD4060 (IC1), 32.768kHz crystal and a few discrete components to generate 2Hz frequency at pin 3 of CD4060. The 2Hz frequency is divided by '2' using IC CD4017 (IC2) to provide 1Hz output at pin 2 for clock input and also generate 1 Hz at pin 3 for setting the time, date etc. The 1Hz clock frequency from pin 2 is fed to the seconds counter-cum-display circuit as shown in Fig. 4.

Seconds counter and display. It is a simple two-digit counter that displays up to 59 seconds. It is built using two ICs CD4033 (IC3 and IC4), two NOR gates of IC CD4001 (IC5) and two LT543 common-cathode displays (DIS1 and DIS2). The circuit receives 1Hz input from the clock generator. NOR gates N1 and N2 reset the counters at every 60th input pulse, thus providing a pulse-per-minute (PPM) output at pin 5 of IC4. This PPM is used as the input for the minutes counter and display circuit. The counters can be reset to '00' by momentarily pressing push-to-on switch S1.

Minutes counter and display. This circuit consists of decade counter CD4017, which counts the incoming pulses and displays 1, 2, 3 and 4 minutes by glowing LED1 through LED4 at Q1 through Q4 outputs of IC6, respectively. The Q5 output resets IC6 and also provides one pulse every five minutes to IC7 to light up LEDs at its Q1 through Q8 outputs, respectively, at five-minute interval like